

Contribution Statement

ChartPRM: Process Reward Modeling and Step-Level Preference Optimization for Complex Multimodal Chart Reasoning

Process Reward Modelling Seminar • University of Heidelberg • September 20, 2026

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Both team members contributed meaningfully to this project. Responsibilities and completed tasks are listed below.

Yahor Lahunovich

Architected the core ChartPRM pipeline on **CharXiv**, engineered custom VLM alignment trainers, and authored the foundational report.

- Curated the 500-question CharXiv reasoning pool (8 domains, 62 chart types) and established the protected 100-question holdout.
- Designed the structured reasoning protocol for Qwen2.5-VL-3B and evaluated 1,274 rollouts via muse-spark-1.1 with cost-saving rollout batching.
- Conducted semantic error analysis across 2,920 refuted steps, establishing the 9-category failure taxonomy that uncovered the perceptual bottleneck and grounded the reward tree.
- Engineered minimal, native PyTorch training engines for SFT, Full DPO, Step-DPO, KTO, and SFT→DPO, bypassing TRL VLM memory bugs to train within a single 16GB T4 GPU envelope.
- Benchmarked all alignment variants on the holdout, diagnosing and resolving policy log-prob collapse and formatting trade-offs.
- Built the centralized scientific visualization framework (SciencePlots, Paul Tol palettes) generating all 29 benchmark and diagnostic figures.
- Set up the ACL LaTeX layout, authored the Introduction, Related Work, ChartPRM Methods, and Results & Analysis sections, and curated the bibliography.
- **AI usage summary:** Used **Antigravity** and **Cursor** (powered by Gemini and Claude) as AI coding assistants for scaffolding scripts, resolving CUDA/VRAM driver bugs on Kaggle, and refactoring pipelines. I conceptualized the small-data hypothesis, designed the experimental protocols, formulated the analytical narratives, and verified all empirical results.

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Presented Yin et al.'s *Dynamic and Generalizable Process Reward Modeling* (DG-PRM) for the seminar and directed its adaptation, plus a cross-dataset study.

- Directed the reward-tree construction: 33 scored criteria under 9 categories, from clustered judge critiques.
- Directed the dynamic multi-criteria judge (scored via gemini-3.5-flash-lite, distinct from the muse-spark-1.1 binary judge) and Pareto-dominance-filtered pair selection for DPO (**Pareto-DPO**).
- Directed the reference-free **SimPO** baseline implementation.
- Directed the ChartGemma specialist-baseline and oracle-ceiling diagnostic experiments.
- Designed and ran the cross-dataset generalization test on **ChartQA**: sampling, rollout generation, unmodified reward-tree transfer, and retraining/evaluating all seven methods on a 30-question holdout.
- Authored the DynamicPRM, Test-Time Verification and Search, and Cross-Dataset Generalization (ChartQA) sections, and their tables/figures; also revised the Abstract, Introduction, and ChartPRM Framework as the project's scope grew to two benchmarks and seven methods.
- **AI usage summary:** Used **Claude Code** (Anthropic) as an AI coding agent for essentially all of the code and analysis above (reward tree, dynamic judge, SimPO, ChartQA pipeline) and for drafting my report sections. I directed every experiment (what to try, when to pivot), set the evidentiary bar (e.g., requiring bootstrap significance tests before claiming a win), and personally verified and interpreted all results before they went into the report.

Both authors confirm this statement accurately reflects their respective contributions to the ChartPRM project.